Document Variables (_vars)

When creating a report yaml document, you have the option to include document-level variables by using the '_vars' key in the root of the document.

These variables can be rendered as part of your strings (using jinja) within the report content using the double-braces (e.g. {{variable_name}}). When rendered using jinja, all values are stringified and only the string representation is shown (i.e. the result of str(var)).

This document has five variables defined, a, b, c, d, e.

- a = hello
- b = True
- c = 23
- d = -0.013
- e = val2
- f = ['val1', 'val2']

Passing Python Objects

It is also possible to pass variable data as Python objects to blocks (see "YMPrint blocks") using the $var syntax (similar to bash).

You cannot use the dollar-sign variable name syntax (e.g. $varname) within content that is to be displayed (if attempted, an error will be raised). The intended use-case is for passing data within the document to be used as *block parameters*.

For example, you might want to have some consistent spacer used throughout your document. Instead of typing "_spacer: 5" at all locations, you can instead set a variable in the _vars to something like "small_space: 5" and employ the variable at each spacer as such "_spacer: $small_space".

The data you set in the _vars key can be any of the native YAML datatypes: str, int, float, null/None, bool, list, and dict.

Computing new Python objects within code

It is also possible to use a _py code block (see "YMPrint blocks") to create variables of arbitrary classes/datatypes and pass that data to block parameters.

A ready example of this would be the _matplotfig block which has a "fig" parameter that expects a matplotlib figure.

You would first generate the figure in a _py block and assign the generated figure to a variable. You would then use the dollar-sign syntax to pass that computed variable into the "fig" parameter of the _matplotfig code block (see YMPrint blocks for details.)